

Homework Classical Physics, Fall 2023

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Problem Set 02.2 (L03 & L04)

Due on Thursday, 21. September 2023, at 09:30 (in the morning!)

(= Deadline for submitting a pdf file with solution)

Mechanics - vectors (4 points)

a) Draw the following vectors in a coordinate system $\mathbf{u} = (2.5, 1.0)$ and $\mathbf{v} = (-0.5, -2.0)$, and plot the sum $\mathbf{v} + \mathbf{u}$ and the difference $\mathbf{u} - \mathbf{v}$. Calculate the dot product $\mathbf{u} \cdot \mathbf{v}$.

b) Calculate the dot product $\vec{a} \cdot \vec{b}$ of two vectors with $\vec{a} = \begin{pmatrix} 2 \\ -4 \\ 1 \end{pmatrix}$, $\vec{b} = \begin{pmatrix} 4 \\ 5 \\ 2 \end{pmatrix}$

c) Calculate the length of vector $\vec{c} = \vec{a} + \vec{b}$ and its angle Θ to the x-axis.

d) Calculate the cross product $\vec{c} = \vec{a} \times \vec{b}$ and the according length of the vector $\vec{c}$